

LogSoftmax#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.LogSoftmax.html>

Description:

Applies the $\log(\text{Softmax}(x))$ function to an n-dimensional input Tensor. The LogSoftmax formulation can be simplified as: Input: $(*)^{(*)} (*)$ where $*$ means, any number of additional dimensions Output: $(*)^{(*)} (*)$, same shape as the input dim (int) – A dimension along which LogSoftmax will be computed.

Function Signature

```
class torch.nn.modules.activation.LogSoftmax(dim=None)
[source]#
```

Shape:

Parameters

Returns

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.LogSoftmax(dim=1)
>>> input = torch.randn(2, 3)
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies the $\log(\text{Softmax}(x))$ function to an n-dimensional input Tensor.

The LogSoftmax formulation can be simplified as:

Input: $(*)^{(*)} (*)$ where $*$ means, any number of additional dimensions

Output: $(*)^{(*)} (*)$, same shape as the input

dim (int) – A dimension along which LogSoftmax will be computed.

a Tensor of the same dimension and shape as the input with values in the range $[-\infty, 0)$

Return the extra representation of the module.

Runs the forward pass.